

CNC foam laser cutter Machine

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Abstract

Computer Numerical Control (CNC) machines have become a fundamental part of modern manufacturing and prototyping due to their ability to produce complex shapes with high accuracy, repeatability, and efficiency. Among their diverse applications, CNC foam cutting machines are particularly valuable in aerospace, packaging, insulation, and education, where lightweight foam materials must be shaped into precise geometries such as airfoils and structural models. Manual foam cutting is time-consuming, imprecise, and inconsistent, which highlights the importance of developing affordable and reliable CNC foam cutters.

This project presents the design and implementation of a CNC foam cutting machine driven by an Arduino microcontroller, TB6600 stepper motor drivers, NEMA stepper motors, and a Nickel-Chromium (NiCr) hot wire as the cutting element. The development followed a structured process: beginning with single-motor control, progressing to synchronized operation of two, three, and four motors, and ultimately achieving full four-axis control. To validate motion control, GRBL firmware was installed on the Arduino to interpret G-code commands, enabling the system to execute programmed tool paths. Initial validation was performed with a pencil to draw squares and airfoil profiles before integrating the NiCr hot wire for actual foam cutting.

Experiments were also conducted to evaluate the heating performance of NiCr wires with different diameters (0.32 mm and 0.6 mm), analyzing the relationship between wire temperature and current. Results demonstrated that thinner wires reach cutting temperatures at lower currents, while thicker wires require higher currents but offer increased durability. Based on these findings, a 0.3 mm NiCr wire was selected as the optimal cutting element.

The final machine successfully cut foam airfoil profiles with high precision, confirming the feasibility of the design. This project demonstrates how low-cost components and open-source tools can be combined to replicate industrial CNC principles, making the machine a practical solution for prototyping, research, and educational applications.

Chapter 1

Introduction

1.1 Background on CNC Machines

Computer Numerical Control (CNC) machines are automated systems that translate digital instructions into precise physical movements. Using programming languages such as G-code, CNC machines eliminate the need for manual cutting, shaping, or drilling, enabling high accuracy, repeatability, and efficiency.

CNC technology has become a cornerstone of modern manufacturing, contributing significantly to Industry 4.0 and digital transformation. It allows for mass production, rapid prototyping, and the fabrication of complex geometries that cannot be easily achieved by hand.

1.2 Types of CNC Machines

CNC systems are versatile and applied across multiple industries. Common types include:

- CNC Milling Machines and Machining Centers – Used for cutting, drilling, and shaping metals and alloys in aerospace, automotive, and industrial manufacturing.
- CNC Routers – Designed for cutting wood, plastics, composites, and lightweight metals, widely used in sign-making and furniture production.
- CNC Laser Cutters – Employ a focused laser beam to cut or engrave materials like wood, plastics, foam, or thin metals.
- CNC Plasma Cutters – Use ionized gas plasma to cut conductive metals, particularly useful in sheet metal fabrication.
- CNC Waterjet Cutters – Utilize high-pressure water (sometimes mixed with abrasives) to cut a wide range of materials without thermal damage.
- CNC 3D Printers – Follow additive manufacturing principles, building parts layer by layer rather than removing material.

Each machine type is optimized for specific materials and applications, but all share the same fundamental principle: computer-controlled precision.



Fig1.1:CNC Milling Machines

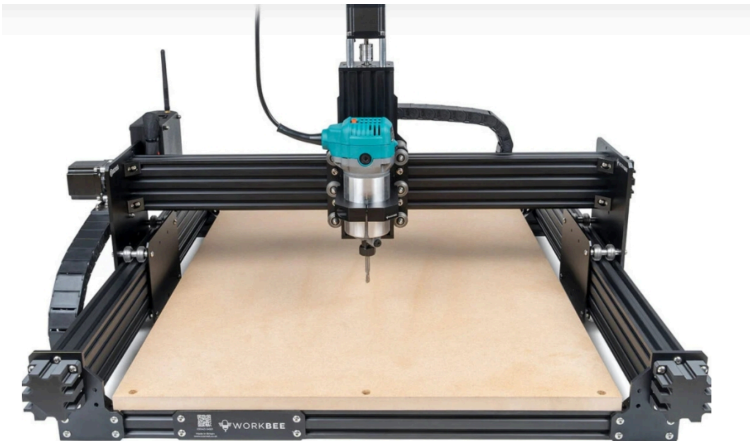


Fig1.2: CNC Routers



Fig1.3:CNC Laser Cutters

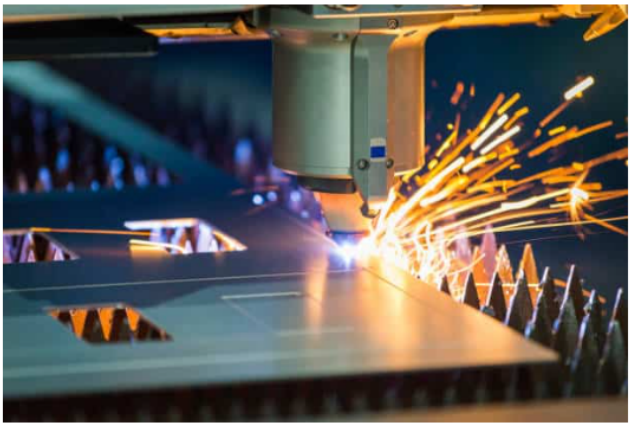


Fig1.4: CNC Laser Cutters



Fig1.5:CNC Plasma Cutters

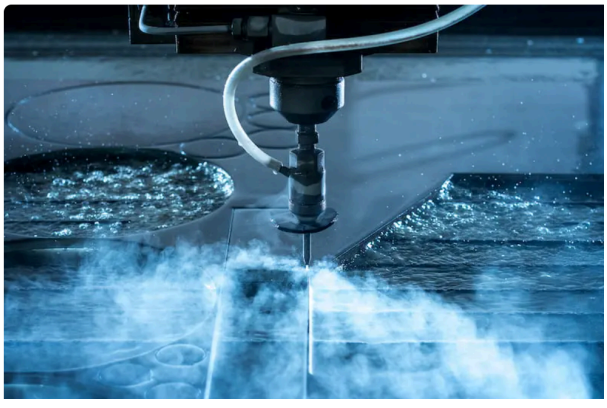


Fig1.6: CNC Waterjet Cutters

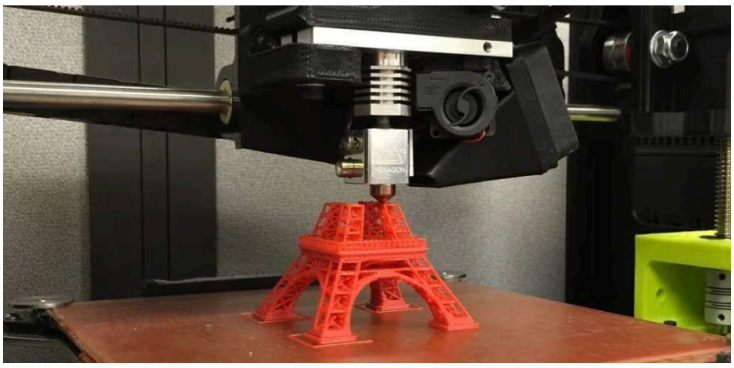


Fig1.7:CNC 3D Printers

1.3 Applications of CNC Technology

CNC machines have transformed modern engineering and production. Their main applications include:

- Aerospace – Production of precision airfoils, turbine blades, and lightweight structural components.
- Automotive – High-accuracy parts such as engine blocks, gear systems, and custom prototypes.
- Packaging and Insulation – Foam cutting for protective packaging inserts and thermal insulation panels.
- Architecture and Design – Creating decorative elements, prototypes, and scaled models.
- Education and Research – Providing practical training for students in automation, robotics, and digital fabrication.

1.4 CNC Foam Cutting Machines

Within this broad field, CNC foam cutting machines specialize in shaping foam materials with precision. Foam is lightweight, inexpensive, and easy to process, making it ideal for:

- Aerospace prototyping – Creating wing and airfoil models for aerodynamic testing.
- Packaging – Custom foam inserts for electronics, medical devices, and fragile equipment.

- Insulation and Construction – Producing panels and structural models.
- Educational use – Training students in CNC operation and design-to-production workflows.

Instead of cutting with a solid blade, foam cutters typically use a heated Nickel-Chromium (NiCr) wire or, in some designs, a laser beam. The heated wire melts the foam cleanly along the cutting path, producing smooth and accurate results.

1.5 Importance of CNC Foam Cutters

CNC foam cutters are important for several reasons:

1. Precision and Accuracy – They ensure consistency across repeated cuts, which is vital for aerospace and engineering applications.
2. Efficiency – Automated cutting reduces time and labor compared to manual methods.
3. Complex Shape Capability – Intricate geometries like airfoils, curved patterns, or customized packaging are easily produced.
4. Low-Cost Prototyping – Foam is an inexpensive material, making it suitable for early-stage model testing.
5. Educational Value – Such machines provide practical exposure to CNC concepts, bridging theory and practice for students and researchers.

1.6 Aim of This Project

The primary aim of this project was to design and construct a CNC foam cutting machine using:

- An Arduino microcontroller for control.
- TB6600 stepper motor drivers for motor power and precision.
- NEMA stepper motors to achieve controlled motion across four axes.
- A Nickel-Chromium (NiCr) hot wire as the cutting element.

The machine was developed step by step, beginning with single-motor control and scaling up to four-motor synchronized operation. After integrating GRBL firmware and G-code control, the system was tested by drawing and later cutting airfoil shapes.

1.7 Conclusion

In this chapter, the fundamental concepts of CNC technology were introduced, beginning with its background and significance in modern manufacturing. CNC machines were shown to be central to Industry 4.0, enabling mass production, rapid prototyping, and the fabrication of complex geometries with high precision and repeatability. A wide range of CNC machine types was discussed, each optimized for specific materials and applications but all sharing the same principle of computer-controlled precision.

The applications of CNC technology were highlighted across industries such as aerospace, automotive, packaging, architecture, and education, demonstrating its versatility and global importance. Within this context, CNC foam cutting machines were presented as a specialized application, particularly valuable in aerospace prototyping, packaging, insulation, and training. By employing methods such as hot-wire cutting with Nickel-Chromium (NiCr) wire, these machines provide a cost-effective, accurate, and efficient way to shape lightweight foam materials into precise geometries such as airfoils and structural models.

This background establishes the importance of the present project: the design and implementation of a CNC foam cutting machine. The next chapter will focus on the system overview and hardware components that form the foundation of the machine, including the microcontroller, stepper motors, drivers, power supply, and cutting element.

Chapter 2

System Overview and Components

2.1 System Overview

The CNC foam cutting machine developed in this project is a four-axis system designed to cut lightweight foam materials with precision. It combines electromechanical motion control with thermal cutting technology using a heated Nickel-Chromium (NiCr) wire. The system is driven by an Arduino microcontroller that interprets motion commands and coordinates the movement of four NEMA stepper motors via TB6600 stepper motor drivers.

The design emphasizes modularity, affordability, and scalability. Each motor is independently controlled through a dedicated driver, but all motors are synchronized through G-code commands executed by the Arduino running GRBL firmware. Power is supplied through a regulated DC power source, and grounding practices are carefully implemented to ensure stability and prevent miscommunication between drivers and the controller.

The machine's core functionality can be summarized as follows:

1. Controller – Arduino Mega/Uno running GRBL for G-code interpretation.
2. Drivers – TB6600 drivers providing sufficient current and microstepping for smooth motor operation.
3. Motors – Four NEMA stepper motors driving the cutting frame.
4. Cutting Element – NiCr wire heated by electrical current to melt and cut foam precisely.
5. Power and Wiring – Regulated supply, common ground bus, and DIP switch configurations for current/microstepping.

2.2 Arduino Microcontroller

The Arduino Mega and Arduino Uno were both used at different stages of development.

- Arduino Uno: Used during the initial setup for one and two motors due to its simplicity and ease of programming.

- Arduino Mega: Chosen for the final four-axis system because of its greater number of input/output pins, which allows for independent control of multiple drivers.

The Arduino serves as the brain of the CNC system, generating pulse (PUL), direction (DIR), and enable (ENA) signals that are transmitted to the drivers. When GRBL firmware is installed, the Arduino is capable of interpreting G-code commands directly from a computer, transforming design files into motor movements.

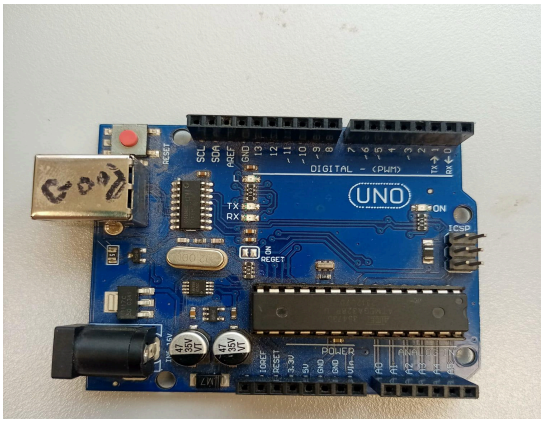


Fig 2.1:Arduino UNO



Fig 2.2:Arduino Mega

2.3 Stepper Motors

The machine employs NEMA 17 stepper motors, rated at 12 V, 0.5 A. These motors were chosen because they:

- Provide sufficient torque for foam cutting, where mechanical resistance is relatively low.
- Offer high precision, as stepper motors move in discrete steps that can be further subdivided by microstepping.
- Are widely available, reliable, and cost-effective.

Stepper motors were mounted on both the X-axis and Y-axis, with each axis driven by two motors for symmetry and stability. This ensured that the cutting wire remains aligned and tensioned during operation.

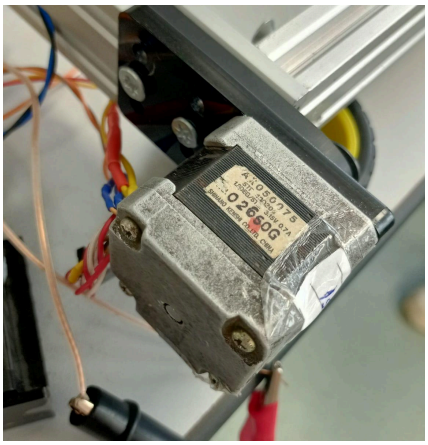


Fig 2.3: Stepper Motor

2.4 Stepper Motor Drivers (TB6600)

The TB6600 driver was selected as the primary motor driver due to its ability to supply up to 4 A per phase and its support for microstepping. Compared to the L298 drivers initially tested, the TB6600 provides:

- Higher current capacity for NEMA 17 motors.
- Microstepping control (up to 1/32) for smooth motion.
- MOSFET-based efficiency with reduced heat generation.
- DIP switch configuration, allowing easy adjustment of current and step resolution.

Each motor has a dedicated TB6600 driver, and the wiring includes PUL+, DIR+, and ENA+ connections to Arduino pins, with PUL-, DIR-, and ENA- tied together to a common ground.



Fig 2.3: Stepper Motor Drivers (TB6600)

2.5 Power Supply

A 12 V DC regulated power supply was used to power both the stepper motors and the NiCr wire. The power supply was selected based on the following requirements:

- Stable voltage to ensure reliable motor operation.
- Sufficient current rating to simultaneously drive four stepper motors.
- Overload protection to safeguard against wiring faults or overheating.

The NiCr cutting wire was powered through the same supply but regulated to ensure appropriate heating without damaging the wire.

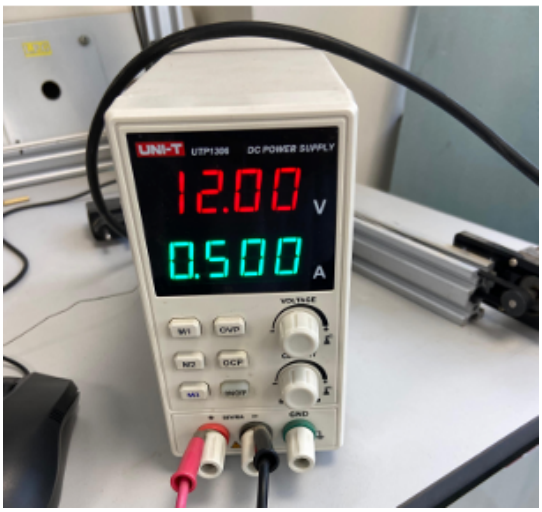


Fig 2.4: Power Supply

2.6 Cutting Element: Nickel-Chromium (NiCr) Wire

The NiCr hot wire is the essential cutting element of the machine. Its properties include:

- High electrical resistivity, allowing it to heat up quickly when current flows.
- High melting point, enabling operation at elevated temperatures without failure.
- Mechanical durability, ensuring stability during cutting operations.

After experimental testing of different diameters, a 0.3 mm NiCr wire was selected as the optimal size, balancing heating efficiency with strength.

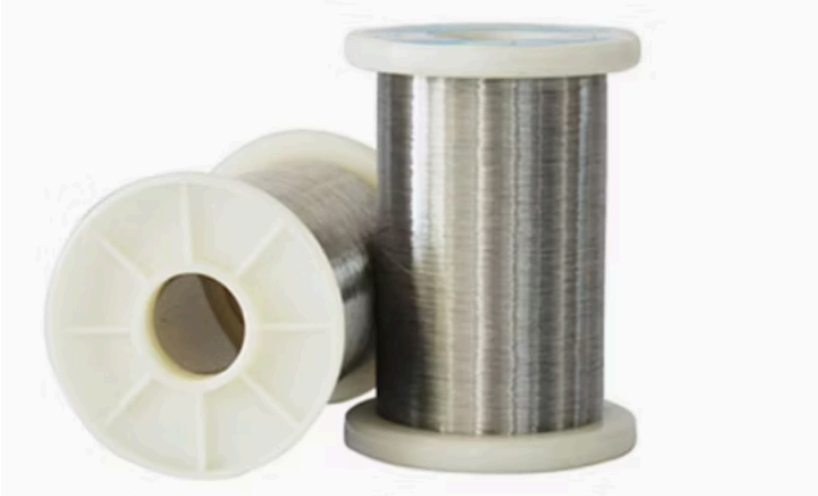


Fig 2.5: Nickel-Chromium (NiCr) Wire

2.7 Prototyping Tools and Accessories

Additional components used in the system include:

- Breadboards and jumper wires for testing connections.
- Switches and connectors for enabling/disabling drivers.
- Foam test materials for validating cutting performance.
- PC interface (USB connection) for uploading G-code to the Arduino.

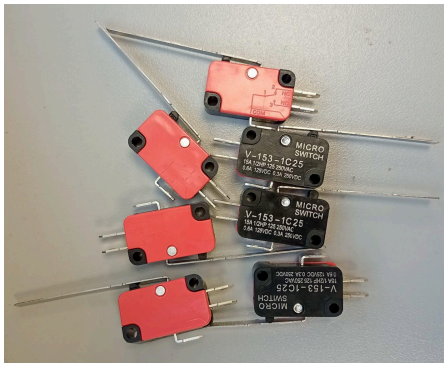


Fig 2.6: Limit Switches



Fig 2.7: Foam test

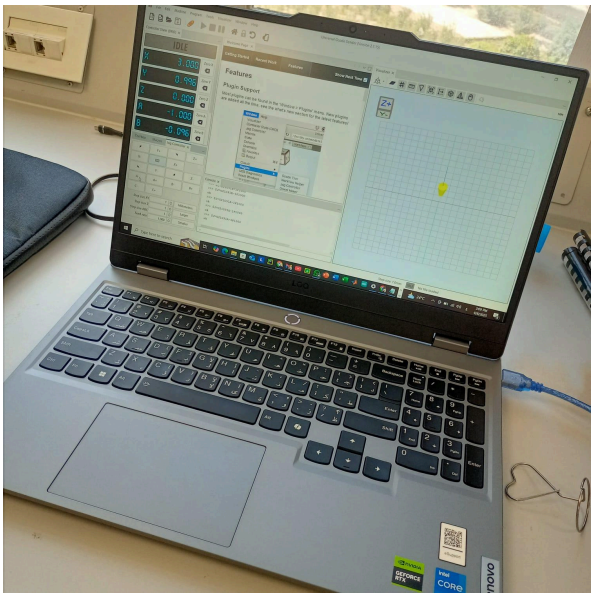


Fig 2.8: PC interface G-code to the Arduino

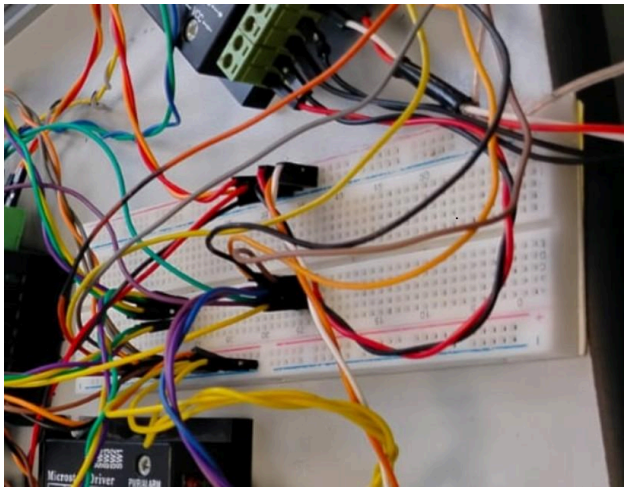


Fig 2.8: Breadboards and jumper wires

2.8 Summary

This chapter presented the main components of the CNC foam cutter, including the Arduino controller, stepper motors, TB6600 drivers, power supply, and NiCr cutting wire. Together, these components form the foundation of the system, enabling precise, programmable cutting of foam materials.

The next chapter will discuss the electrical design and wiring, detailing how each component was interconnected and configured to achieve reliable four-axis control.

Chapter 3

Electrical Design & Wiring

3.1 Introduction

The electrical design of the CNC foam cutting machine forms the backbone of its functionality. It ensures that the Arduino microcontroller, TB6600 stepper motor drivers, stepper motors, and Nickel-Chromium cutting wire operate in synchronization. Proper wiring is essential for achieving stable signals, preventing noise or miscommunication, and providing sufficient power to all components.

This chapter presents the wiring methods used for the machine, including single-motor testing, two-motor configurations, and the final four-motor arrangement. Special attention is given to common grounding, DIP switch configuration, and power distribution to guarantee reliable performance.

3.2 System Development Process

Methodology:

Wiring Diagram:

A bipolar stepper motor with six leads was used. According to the datasheet, the appropriate configuration is the Bipolar Drive Series Connection, using Color Code 2:

- Brown, Black, Orange, Red, White, Yellow
- A coil: Brown (A+), Orange (A−)
- B coil: Red (B+), Yellow (B−) The following diagram shows the wiring logic:

6 LEAD WIRES	1	2	3	4	5	6
Color Code 1	Red	White	Blue	Green	Yellow	Black
Color Code 2	Brown	Black	Orange	Red	White	Yellow
Color Code 3	Red	Black	Red / White Stripe	Green	White	Green / White Stripe
Bipolar Drive Half Coil Connection	A	$\overline{A}$		B	$\overline{B}$	
		$\overline{A}$	A		$\overline{B}$	B
Bipolar Drive Series Connection	A		$\overline{A}$	B		$\overline{B}$
Unipolar Drive	A	A/C Comm	C	B	B/D Comm	D

Fig3.1:Wiring Color Code for Bipolar Stepper Motors

3.2 Stepper Motor Wiring and Color Coding

The NEMA 17 stepper motors used in this project are bipolar stepper motors, each with two coils. Correct wiring of these coils is essential to ensure proper motor rotation.

- Coil A: A+ and A− terminals
- Coil B: B+ and B− terminals

In the motors available in the lab, the color coding followed this convention:

- Coil A → Red (A+) and Blue (A−)
- Coil B → Green (B+) and Black (B−)

Each pair of coil wires was connected to the corresponding output terminals on the TB6600 driver. Incorrect wiring of coils can cause the motor to vibrate without rotating.

Wiring color code for bipolar stepper motors)

3.3 TB6600 Driver Signal Wiring

Each TB6600 driver requires three key input signals from the Arduino:

- PUL (Pulse) → controls the step signal to move the motor.
- DIR (Direction) → controls motor rotation direction.
- ENA (Enable) → activates or deactivates the driver.

For every driver:

- PUL+, DIR+, ENA+ were connected to assigned Arduino pins.
- PUL−, DIR−, ENA− were tied together across all drivers to form a common ground.
- A single jumper wire connected this common ground to the Arduino GND.

This grounding method ensured that all drivers shared the same electrical reference, preventing noise or loss of steps.

3.4 One-Motor Wiring

For initial testing, a single NEMA 17 stepper motor was connected to one TB6600 driver. The Arduino sent step and direction signals to the driver, while the motor coils were connected to the driver's A+/A− and B+/B− outputs.

This configuration allowed testing of:

- Forward rotation at different speeds.
- Reversing direction.
- Stopping and restarting without error.

Table 1: Driver to Arduino Connections

Signal (Driver Side)	Connection (Arduino UNO)	Wire Color
ENA- (ENA)	Ground	Blue Wire
ENA+ (+5V)	No connection	–
DIR- (DIR)	Ground	Grey Wire
DIR+ (+5V)	Pin 4	Brown Wire
PUL- (PUL)	Ground	Green Wire
PUL+ (+5V)	Pin 3	White Wire

Table 2: Stepper Motor and Power Supply to Driver

Signal (Driver Side)	Connection	Wire Color
A+	Stepper Motor Coil A	Brown Wire
A-	Stepper Motor Coil A	Orange Wire
B+	Stepper Motor Coil B	Red Wire
B-	Stepper Motor Coil B	Yellow Wire
GND	Power Supply Ground	Black Wire
VCC	Power Supply +12V	Red Wire

3.5 Two-Motor Wiring

To control two motors, two TB6600 drivers were connected to the Arduino with separate PUL, DIR, and ENA pins. This allowed either synchronized motion (same direction and speed) or independent motion (different speeds/directions).

Example Arduino connections:

- Motor 1 → PUL+ to pin 7, DIR+ to pin 6, ENA+ to pin 5
- Motor 2 → PUL+ to pin 10, DIR+ to pin 9, ENA+ to pin 8

This is the connection from each driver to the Arduino UNO :

Axis	Signal	TB6600 Pin	Arduino uno Pin
X-Axis	PUL+	Step Input	Pin 7
	DIR+	Direction Input	Pin 6
	ENA+	Enable Input	Pin 5
Y-Axis	PUL+	Step Input	Pin 10
	DIR+	Direction Input	Pin 9
1	ENA+	Enable Input	Pin 8

High Voltage	B-	Yellow Wire	**Stepper to Driver**
	B+	Red Wire	
	A-	Orange Wire	
	A+	Brown Wire	
	GND	Power Supply (Black Wire)	
	VCC	Power Supply (Red Wire)	

Common Grounding Setup

For stable and reliable signal communication between the Arduino Mega and the TB6600 drivers:

- The PUL–, DIR–, and ENA– pins from the two drivers were connected together on a common ground rail of a breadboard.

- From this common ground, a single jumper wire was taken to connect to the GND pin of the Arduino UNO.

3.6 Four-Motor Wiring (Final Configuration)

In the final design, four TB6600 drivers controlled four stepper motors for the X, Y, A, and B axes. Each motor had independent pins assigned on the Arduino Mega.

Axis	Signal	Arduino Pin
X-Axis	PUL+ → A0, DIR+ → A1, ENA+ → 38	
Y-Axis	PUL+ → A6, DIR+ → A7, ENA+ → A3	
A-Axis	PUL+ → 26, DIR+ → 28, ENA+ → 24	
B-Axis	PUL+ → 36, DIR+ → 34, ENA+ → 30	

- All PUL−, DIR−, and ENA− pins were tied together into a common ground rail.
- This common ground was linked to the Arduino Mega GND with a single jumper.
- The power supply's positive terminal was distributed to each driver's VCC input.

This wiring enabled synchronized four-axis motion and independent axis control when programmed through GRBL.

3.7 Power Supply Distribution

A regulated 12 V DC power supply was used for both the motors and the NiCr hot wire. To prevent voltage drops and ensure stable operation:

- The positive terminal was distributed to all TB6600 drivers through a common VCC bus.
- Grounds were unified through the breadboard method.
- The NiCr cutting wire was connected separately, with current regulated according to wire diameter.

3.8 DIP Switch Configuration on TB6600

The TB6600 driver includes DIP switches for setting current limits and microstepping resolution.

- Current was set to match the rated current of the NEMA 17 stepper motors (0.5 A) to prevent overheating.
- Microstepping was configured to balance smoothness and speed, ensuring accurate foam cutting without excessive computation time.

3.9 Summary

The electrical design of the CNC foam cutter was built step by step, starting with single-motor wiring and progressing to a four-motor synchronized system. Key design decisions included:

- Common grounding for all drivers to prevent noise.
- Independent driver connections for precise multi-axis control.
- Power supply distribution for stable voltage and current delivery.
- DIP switch configuration for optimized motor performance.

This wiring foundation ensured reliable motor control, enabling the integration of GRBL firmware and G-code in the next development stage.

Teasting

1.Stepper Motor Driver Selection and Testing:

1.1 Overview

The performance, precision, and reliability of a CNC foam-cutting machine depend significantly on the quality of the motor driver used to control its stepper motors.
During the development phase, two motor driver modules were examined: the L298 dual H-bridge driver and the TB6600 stepper motor driver.
This section presents a detailed comparison between these two drivers, the experimental results obtained during testing, and the rationale for selecting the TB6600 as the final driver for the CNC system.

1.2 Comparison Between L298 and TB6600 Drivers

1.2.1 Type of Motor Controlled

- **L298 Driver:**
  The L298 is a general-purpose H-bridge driver originally designed for controlling DC motors or very small bipolar stepper motors.
  It uses bipolar transistors to switch current and is mainly suited for low-current applications such as robotics, small actuators, or educational projects.

- **TB6600 Driver:**
  The TB6600 is a dedicated stepper motor driver specifically designed to control bipolar stepper motors, such as NEMA 17, NEMA 23, and higher torque types used in CNC machines.
  It operates using Step/Direction (STEP/DIR) signals, allowing it to interface directly with CNC controllers such as Arduino GRBL or CNC Shields for precise motion control.

1.2.2 Limitations of the L298 Driver

Through both theoretical evaluation and lab testing, several limitations of the L298 were observed:

1. **Designed for DC motors:**
   The L298 was never intended for high-torque stepper motor applications. It lacks the advanced features required for precise multi-phase control.

2. **Low current capability:**
   Each channel of the L298 can supply only about 2 A peak, and in practice, the

continuous current is limited to 1–1.5 A due to thermal dissipation.
This is insufficient for driving NEMA stepper motors used in CNC systems, which typically require 2–4 A per phase.

3. **No microstepping support:**
   The L298 cannot generate microsteps or respond to step/dir signals. As a result, motor movement is rough, noisy, and inaccurate, unsuitable for fine positioning required in foam cutting.

4. **Inefficient operation:**
   The L298 is based on bipolar transistors, which have a high voltage drop and low efficiency.
   Significant power is dissipated as heat, requiring large heatsinks and causing instability during continuous operation.

1.3 Laboratory Testing of the L298 Driver

In the lab, the L298 driver was tested with **NEMA** stepper motors and AC motors to evaluate its performance under real **CNC** conditions.

- **Stepper Motor Test:**
  Only one of the tested L298 modules could rotate a NEMA stepper motor, and even then, the motion was unstable and inconsistent.
  The motor exhibited jerky movements and incomplete steps, indicating insufficient current and lack of microstepping control.

- **AC Motor Test:**
  Attempts to operate AC motors with the L298 driver failed completely, as the driver architecture is not designed for alternating current applications.

- **Thermal Observation:**
  The driver became extremely hot within minutes of operation, confirming its inefficiency and unsuitability for continuous duty cycles.

Conclusion:
The experimental results confirmed that the L298 driver can only be used for simple demonstrations or low-power educational projects, not for industrial or CNC-grade stepper motor control.

1.4 Advantages of the TB6600 Driver

The TB6600 driver was adopted as the final choice for the CNC foam-cutting machine due to its superior electrical performance, control features, and reliability.

Key Advantages:

1. **High Current Capacity:**
   The TB6600 can deliver up to 4.5 A per phase, which is ideal for NEMA 17/23 stepper motors used in CNC applications.

2. **Microstepping Control:**
   It supports multiple microstepping modes (from full-step up to 1/32 microstep), allowing smooth, quiet, and precise motion — crucial for generating accurate airfoil shapes.

3. **Step/Dir Compatibility:**
   The driver interfaces easily with Arduino GRBL or similar motion controllers through step and direction signals, ensuring accurate synchronization between software and hardware motion.

4. **Efficient Operation:**
   It uses MOSFET switching technology, which is more efficient than the bipolar transistors used in L298.
   This results in lower heat dissipation, higher torque output, and greater energy efficiency.

5. **Built-in Protections:**
   The TB6600 includes overcurrent, overtemperature, and short-circuit protection, ensuring stable and safe long-term operation under CNC conditions.

2. Hot-Wire Testing

2.1 Overview

After completing the mechanical assembly and motor-driver validation, the next stage focused on testing the hot-wire cutting subsystem, which is responsible for generating the required heat to melt and shape the foam accurately.
The goal of this phase was to determine the most suitable hot-wire material, diameter, and operating parameters (voltage, current, and feed rate) that could achieve smooth, precise, and repeatable cuts on various Rockal XPS foams.

2.2 Preliminary Machine Verification

Before introducing the heating wire, the CNC machine was first verified mechanically to ensure that the motion control system and the Arduino GRBL firmware operated correctly.

- **First Iteration – Motion Without Hot Wire:**
  The initial test involved running the CNC machine without any heating element.
  A pencil was mounted in place of the wire to simulate the cutting path and verify the X- and Y-axis synchronization.
  During this stage, the machine successfully drew a perfect square, confirming that the stepper-motor control, direction signals, and feed-rate commands were functioning accurately.

- **Airfoil Drawing Verification:**
  After confirming square motion, multiple airfoil profiles were drawn using the same pencil setup.
  This allowed the team to confirm path accuracy, curve smoothness, and dimensional calibration before any thermal components were energized.
  Once the path precision was validated, the system was ready for thermal testing with the actual NiCr hot wire.

2.3 Hot-Wire Selection Iterations

To determine the optimal heating performance, several iterations were conducted using Nickel-Chromium (NiCr) wires of different diameters and voltage-current settings.
The objective was to find the wire size that produced uniform heating, stable cutting, and clean surface finish across the full cutting span.

Iteration 1 – Wire Diameter 0.5 mm

- **Feed Rate: 72 mm/min**
- **Supply Voltage: 13.3 V**
- **Current: Open-loop limit 6 A (max available; machine draws only what it needs)**

At this setting, the 0.5 mm wire heated quickly and was able to cut through the XPS foam, but the wire exhibited slight temperature variation along its length.
This caused minor non-uniform melting near the wire ends, especially in longer cuts.
The wire's smaller cross-section also made it more fragile under tension, requiring careful handling.

Iteration 2 – Wire Diameter 0.6 mm

- **Feed Rate: 72 mm/min**

- **Supply Voltage: 13.5 V**
- **Current: Open to 6 A (max)**

Increasing the diameter improved mechanical stability and provided more uniform temperature distribution along the span.
However, the thicker wire required slightly higher voltage to reach the optimal cutting temperature and responded more slowly to thermal changes.
At this stage, cutting performance improved significantly with smoother edges and less localized melting, particularly on Rockal Foam Plus (32–36 kg/m³).

Iteration 3 – Optimization Run

- Wire Diameter: 0.6 mm
- Feed Rate: 72 mm/min
- Supply Voltage: 12.5 V (optimized setting)

This iteration produced the best cutting performance.
The wire maintained a stable red-orange glow, indicating uniform heat, while foam surfaces showed minimal kerf width, no discoloration, and excellent smoothness.
The current automatically stabilized below 6 A as the foam melted cleanly with consistent resistance.
This operating point (feed rate 72 mm/min, 12.5 V, NiCr 0.6 mm) was selected as the final configuration for subsequent production airfoil cuts.

2.4 Observations and Evaluation

- The 0.5 mm wire delivered acceptable cutting but was more fragile and prone to temperature variation.
- The 0.6 mm wire demonstrated superior uniformity and long-term stability under CNC tension.
- Optimal results occurred at 12.5 V supply with a feed rate of 72 mm/min, where the foam melted uniformly without smoke or surface deformation.
- The system's current-control loop successfully allowed the wire to draw the exact power needed for different foam densities, improving safety and efficiency